Julian Cutaya

Second-year engineering student at ENSEEIHT, HPC and Big Data track

Seeking a 10-week to 3-month internship in AI or data science, June to mid-September 2027

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Education

ENSEEIHT (Toulouse INP), Master of Engineering (Diplôme d'ingénieur) 2026 – 2028
Digital Sciences, High-Performance Computing and Big Data track Toulouse

Double Bachelor's degree in Mathematics and Computer Science 2023 – 2026
Université de Toulouse Toulouse

French Baccalauréat, highest honours (mention Très Bien), Lycée Pierre-Paul Riquet 2023

Research and projects

Supervised research project, IRIT laboratory 2025 – 2026
Limits of an online time-series clustering algorithm, supervised by Emmanuelle Claeys Toulouse

- Studied StreamingDBAKMeans, an online k-means for time series (DTW distance, DBA barycenters)
- Designed a synthetic data generator and a reproducible benchmarking protocol
- Analysed the effect of noise, class separability, class imbalance and hyperparameters on accuracy

Human motion analysis and prediction from video 2025 – 2026
Bachelor's project, Python and MMPose

- Frame-by-frame 2D pose estimation to extract the skeleton of the person on screen
- Kinematic analysis of the joints, then prediction of upcoming motion with an LSTM network

Enedis hackathon on missing data, team project 2025

- Topic: fill in electricity consumption curves with missing measurements
- Approach: clustering of consumption profiles, then reconstruction of missing values by linear interpolation and from each cluster's centroid

Mathematics thesis, bachelor's project 2024 – 2025

- Proof of Fermat's theorem on sums of two squares (modular arithmetic, rings)

Strength-training tracker mobile app, personal project Since 2025

- Designed and built solo, from database to interface, with Flutter and Firebase

Experience

Private mathematics tutoring 2024 – 2025
Self-employed, high-school level (10th and 12th grade)

- Tutored two students over several months: reviewing concepts, exercises, test preparation
- Prepared a 12th-grade student for the 2025 baccalauréat maths exam: 9/20 year average $\rightarrow$ 15/20 on the exam

Skills

Fields: machine learning, time series, computer vision, parallel computing

Programming: Python (NumPy, pandas, scikit-learn), C (OpenMP), Java, OCaml, SQL

Tools: Linux, Git and GitHub, Jupyter, VS Code, LaTeX

Languages: French (native), English (TOEIC 940/990), Spanish (B1), Japanese (beginner, second foreign language)

Interests

Badminton for 10 years (school and university sports associations, club), strength training